جامعة أم القرى
UMM AL-QURA UNIVERSITY

Kingdom of Saudi Arabia

Ministry of Education

Umm Al-Qura University

College of Engineering and Computers at Al-Qunfudah

Bachelor of Science in Computer Science

Group Project: Deploying a Static HTML Website Using Amazon S3

Created by:

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Prepared For:

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Cloud Computing Project: Deploying a Static HTML Website Using Amazon S3

1. Personal Information

AYAK **Public Website URL:** https://ayak-website-bucket.s3website.eu-north-1.amazonaws.com/

d1ymgw7t5nrgum.cloudfront.net

ID	Name	Role
444******	Eng. Khaled Mahmoud Sulaimani	Team Leader / AWS Specialist/ Web Developer/ Documentation Specialist

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2. Project Description

This website serves as the digital front page for **"AYAK"**, a tech startup specializing in providing smart, innovative solutions in the Internet of Things (IoT) and embedded systems. Developed using HTML, CSS, and JavaScript, the site is designed to professionally showcase the platform's vision, IoT hardware, and project team through a modern and interactive user experience.

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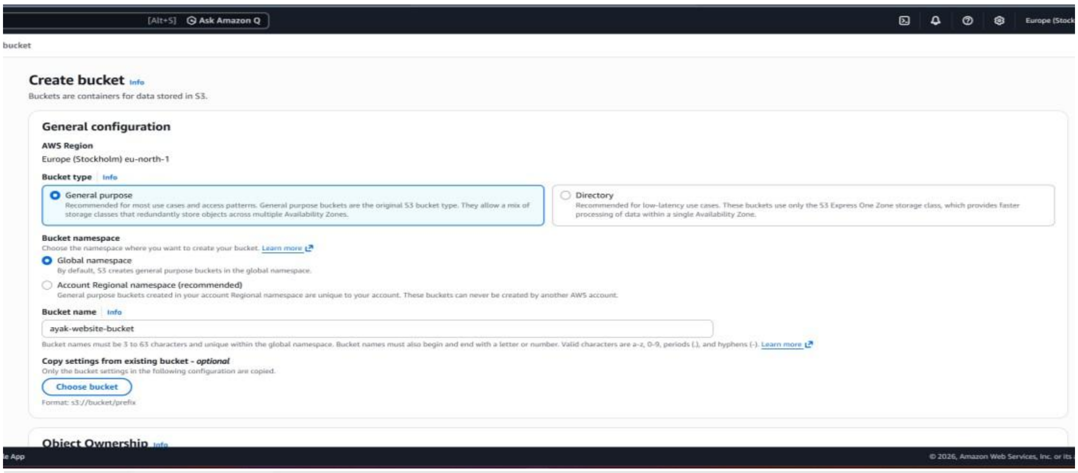
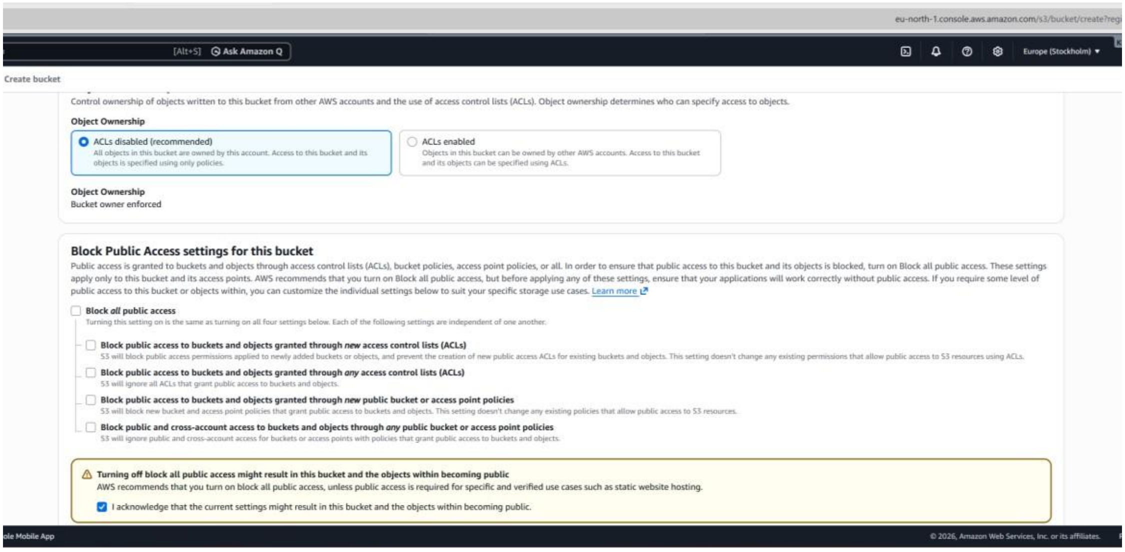
3. Cloud Implementation Steps

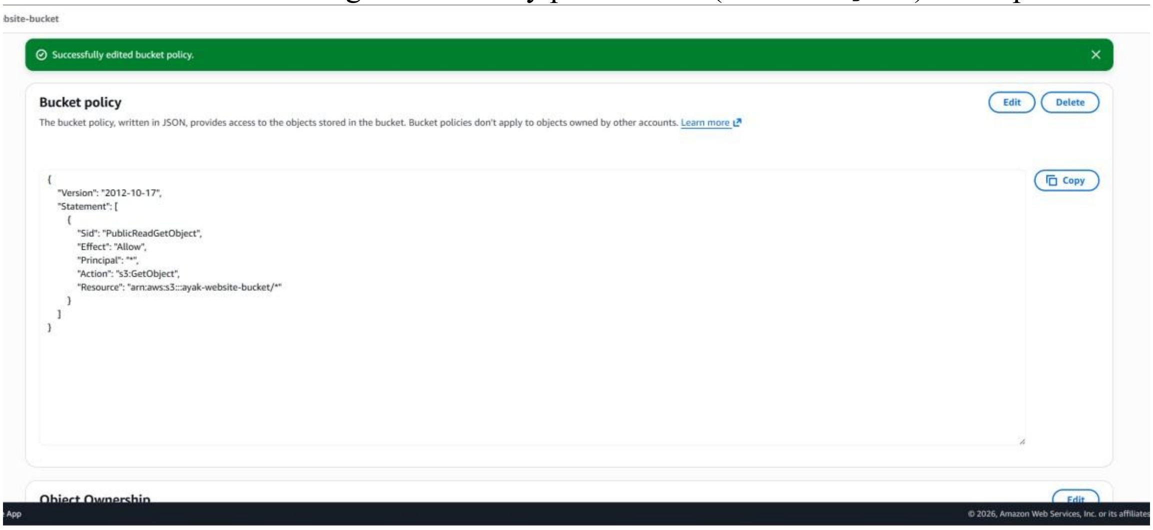
The website files were successfully hosted on a cloud environment using Amazon S3. Below is the documentation of the implementation process:

Step 1: Creating the Storage Space (S3 Bucket) A new bucket was created under the AWS Free Tier with a globally unique name (`ayak-website-bucket`) to serve as the main container for the website files.

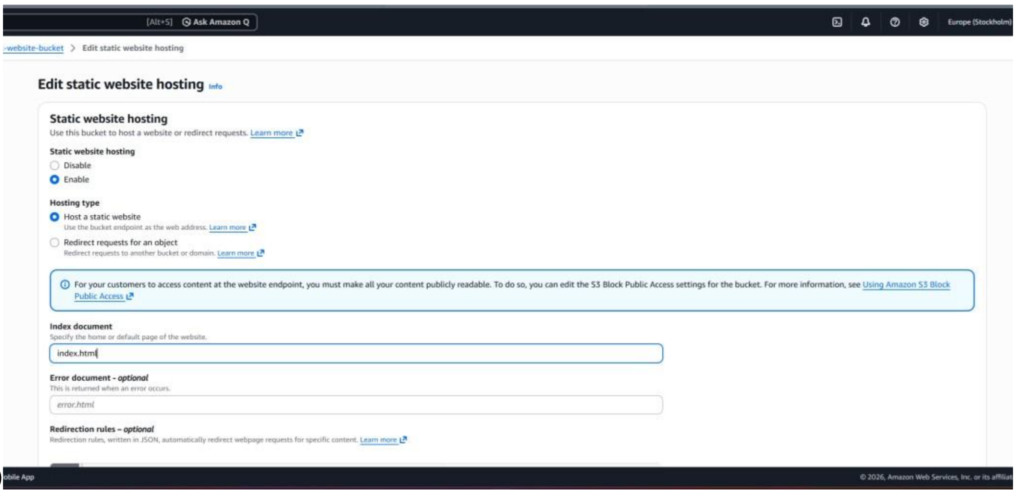




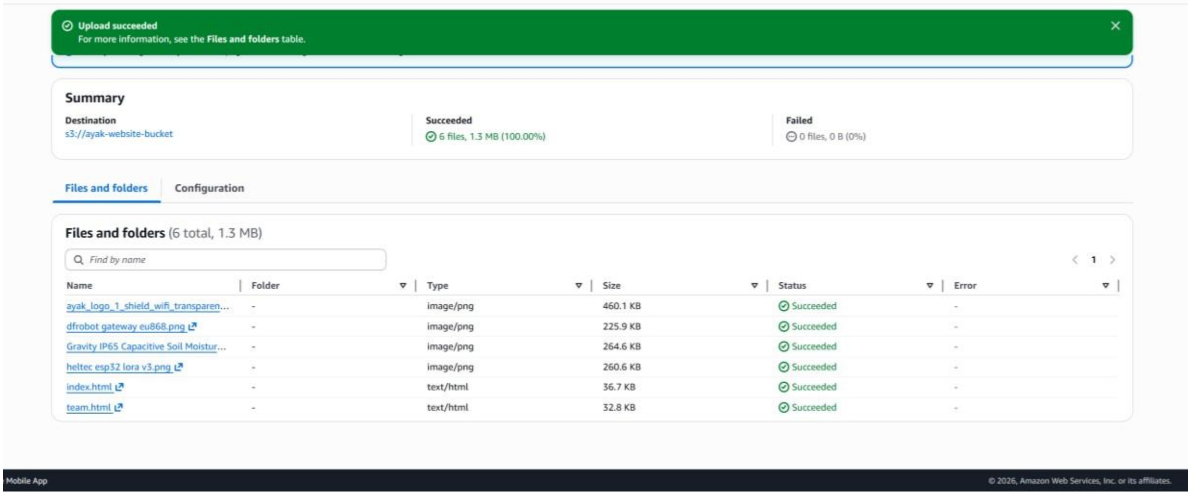
Step 2: Security Settings & Public Access To make the website accessible to visitors, the "Block all public access" setting was disabled. A Bucket Policy written in JSON was then attached to grant read-only permissions (`s3:GetObject`) to the public.



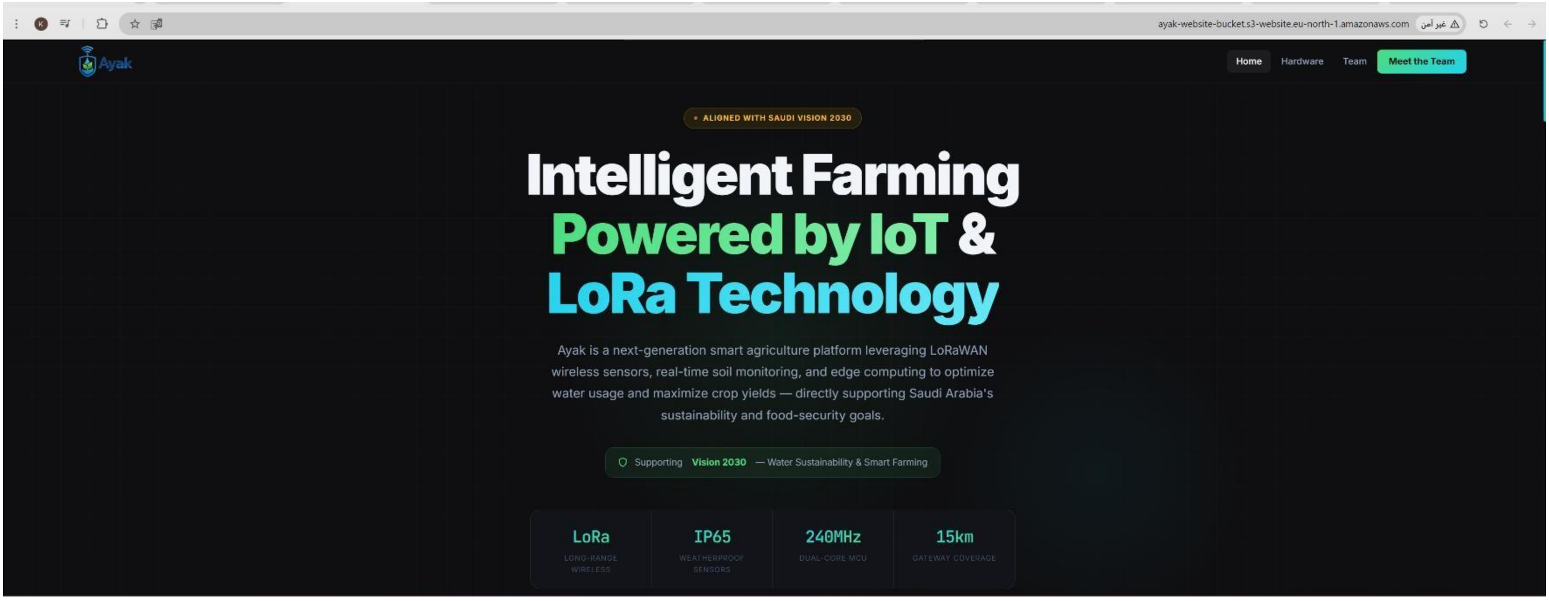
Step 3: Enabling Static Website Hosting The static website hosting feature was enabled from the bucket's properties. The `index.html` file was specified as the default index document to load when visiting the endpoint.



Step 4: Uploading Website Files All project files, including HTML, CSS, and image assets, were successfully uploaded to the S3 bucket.



Step 5: Final Result & Deployment The public endpoint URL was tested via a web browser. The static website loaded successfully and is now accessible to the public.



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4. Security Awareness & S3 Bucket Policies

As part of understanding the shared responsibility model in cloud computing, we analyzed the security implications of hosting a static website on Amazon S3:

- **S3 Bucket Policies:** These are JSON-based security rules used to manage access permissions at the bucket level. For this project, we configured the policy to allow anonymous users (the public) to read objects (`"Action": "s3:GetObject"`) without granting them any permissions to modify, upload, or delete data, which is the standard practice for static web hosting.
- **Security Risks of Enabling Public Access:** Disabling the "Block Public Access" feature carries inherent risks. If sensitive data (such as passwords, API keys, personal user data, or internal company documents) is accidentally uploaded to this public bucket, it will be exposed and accessible to anyone on the internet. Furthermore, unrestricted public access can make the bucket vulnerable to malicious scraping or bot traffic, which can consume significant bandwidth and result in unexpected financial charges on the AWS account.